EXERCISE – 6.3

MULTIPLE LINEAR REGRESSION

1. The following are sample data provided by a moving company on the weights of six shipments, the distances they were moved, and the damage that was incurred:

Weight (1,000 *pounds*) x_1	*Distance* (1,000 *miles*) x_2	*Damage* (*dollars*) y
4.0	1.5	160
3.0	2.2	112
1.6	1.0	69
1.2	2.0	90
3.4	0.8	123
4.8	1.6	186

(a) Fit a multiple linear regression model using the above data set also interpret the results.
(b) Estimate the damage when a shipment weighing 2,400 pounds is moved 1,200 miles.
(c) Find coefficient of determination and interpret it.
(d) Test the parameters of the multiple linear regression model.

$(Ans: (a)\hat{y} = 14.56 + 30.11x_1 + 12.16x_2 \;\; (b)\hat{y} = \$101.41 \;\; (c)R^2 = 0.9197)$

2. The following are data on the average weekly profits (in $1,000) of five restaurants, their seating capacities, and the average daily traffic (in thousands of cars) that passes their locations:

Seating capacity x_1	*Traffic count* x_2	*Weekly net profit* y
120	19	23.8
200	8	24.2
150	12	22.0
180	15	26.2
240	16	33.5

(a) Fit a multiple linear regression model using the above data set also interpret the results.
(b) Predict the average weekly net profit of a restaurant with a seating capacity of 210 at a location where the daily traffic count averages 14,000 cars.
(c) Find Coefficient of Determination and interpret it.
(d) Test the parameters of the multiple linear regression model.

$(Ans: (a)\hat{y} = -0.627 + 0.097x_1 + 0.662x_2 \;\; (b)\hat{y} = 84 \;\; (c)R^2 = 0.9943)$

3. The following data represent the chemistry grades for a random sample of 12 freshmen at a certain college along with their scores on an intelligence test administered while they were still seniors in high school. we were also given the number of class periods missed by the 12 students taking the chemistry course. The complete data are shown.

Student	Chemistry Grade, y	Test Score, x_1	Classes Missed, x_2
1	85	65	1
2	74	50	7
3	76	55	5
4	90	65	2
5	85	55	6
6	87	70	3
7	94	65	2
8	98	70	5
9	81	55	4
10	91	70	3
11	76	50	1
12	74	55	4

(a) Fit a multiple linear regression model using the above data set also interpret the results.
(b) Estimate the chemistry grade for a student who has an intelligence test score of 60 and missed 4 classes.
(c) Find Coefficient of Determination and interpret it.
(d) Test the parameters of the multiple linear regression model.

$(Ans: (a)\hat{y} = 27.5467 + 0.9217x_1 + 0.2842x_2 \;\; (b)\hat{y} = 84 \;\; (c)R^2 = \quad)$

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4. The following are data on the percent effectiveness of a pain reliever and the amounts of three different medications (in milligrams) present in each capsule:

Medication A x_1	*Medication B* x_2	*Medication C* x_3	*Percent effective* y
15	20	10	47
15	20	20	54
15	30	10	58
15	30	20	66
30	20	10	59
30	20	20	67
30	30	10	71
30	30	20	83
45	20	10	72
45	20	20	82
45	30	10	85
45	30	20	94

(a) Fit a multiple linear regression model using the above data set also interpret the results.
(b) Predict percent effectiveness of a pain reliever if a capsule contains Medication A 12 mg, Medication B 25mg and medication C 8mg.
(c) Find Coefficient of Determination and interpret it.
(d) Test the parameters of the multiple linear regression model.

$(Ans:(a)\hat{y}=-102.71+0.61x_1+8.92x_2+1.44x_3+0.01x_4\ (b)\ \hat{y}=287.56\ (c)\ R^2=\quad)$

5. The electric power consumed each month by a chemical plant is thought to be related to the average ambient temperature $x1$, the number of days in the month $x2$, the average product purity $x3$, and the tons of product produced $x4$. The past year's historical data are available and are presented in the following table.

y	x_1	x_2	x_3	x_4
240	25	24	91	100
236	31	21	90	95
290	45	24	88	110
274	60	25	87	88
301	65	25	91	94
316	72	26	94	99
300	80	25	87	97
296	84	25	86	96
267	75	24	88	110
276	60	25	91	105
288	50	25	90	100
261	38	23	89	98

(a) Fit a multiple linear regression model using the above data set also interpret the results.
(b) Predict power consumption for a month in which $x1 = 75$◦F, $x2 = 24$ days, $x3 = 90\%$, and $x4 = 98$ tons.
(c) Find Coefficient of Determination and interpret it.
(d) Test the parameters of the multiple linear regression model.

$(Ans:(a)\hat{y}=-102.71+0.61x_1+8.92x_2+1.44x_3+0.01x_4\ (b)\ \hat{y}=287.56\ (c)\ R^2=\quad)$

Note: In all the Questions Use Matrices Approach for Estimation.

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